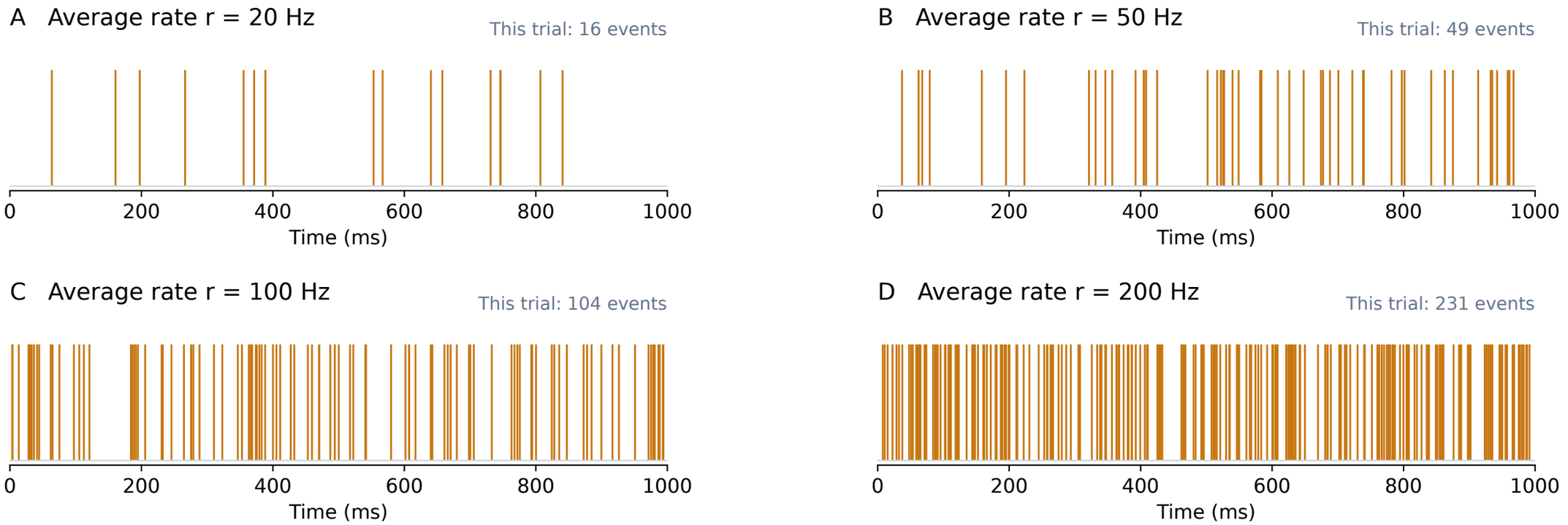
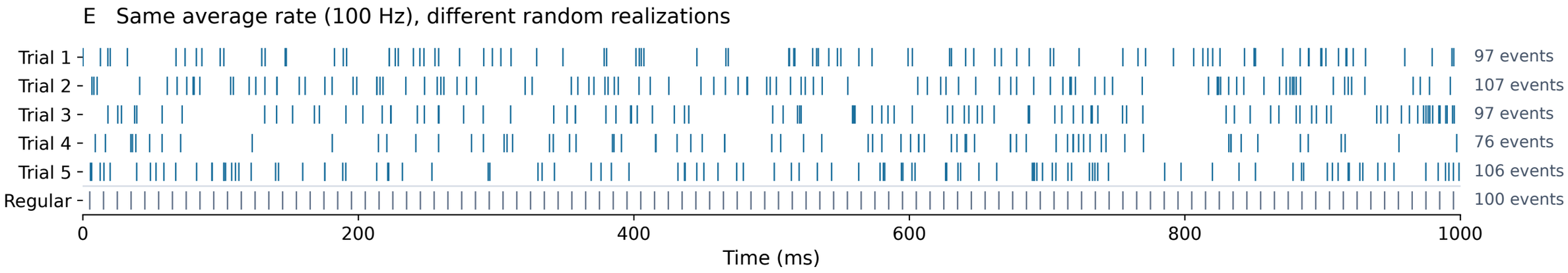
# Poisson input: random event times at a fixed average rate

A rate of 100 Hz means 100 events per second on average; individual intervals and event counts vary.

$N(T) \sim \text{Poisson}(rT)$;  $\Delta t \sim \text{Exponential}(r)$;  $E[\Delta t] = 1/r$  (T in seconds, r in Hz)




Each vertical tick is one event arrival. Tick height is symbolic and does not represent g or voltage.

Poisson at 100 Hz: mean interval = 10 ms. Regular at 100 Hz: every interval = 10 ms. Seed: 20261104.